

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	XXXXX
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal Report

"Raising Waters" is an innovative **AI-powered flood prediction and early warning system** designed to address the growing challenge of flood disasters in India and other flood-prone regions worldwide. The system leverages **Machine Learning algorithms**, **real-time weather data integration**, and a **user-friendly dashboard** to provide accurate flood predictions with a lead time of **48-72 hours**.

Project Overview	
Objective	To design, develop, and implement an AI-powered flood prediction and early warning system that leverages Machine Learning models to analyze real-time weather data and provide accurate, timely flood predictions with automated alerts..
Scope	Machine Learning models to analyze real-time weather data and provide accurate, timely flood predictions with automated alerts..
Problem Statement	
Description	The Raising Waters project is an AI-powered flood prediction and early warning system that integrates real-time weather data from multiple sources (OpenWeatherMap, IMD, and sensors) with advanced machine learning models (LSTM, Random Forest, XGBoost) to deliver highly accurate flood forecasts with 48-72 hours of advance warning through an interactive dashboard and automated multi-channel alert system.

Impact	This system will reduce flood-related casualties by up to 70% and cut economic losses by 50% by enabling timely evacuations and proactive resource allocation. It empowers vulnerable communities, farmers, fishermen, and disaster management teams with actionable, real-time flood risk information, transforming flood response from reactive to proactive, data-driven disaster management..
Proposed Solution	
Approach	The project follows an Agile-Scrum methodology with 2-week sprints, combining incremental development with continuous stakeholder feedback. Our approach integrates data engineering (real-time ingestion from 5+ weather APIs), machine learning (LSTM, Random Forest, XGBoost with ensemble methods), and user-centered design to build a robust, scalable flood prediction system, with automated retraining pipelines ensuring models continuously improve with new data.
Key Features	Real-time flood risk maps with color-coded zones using Leaflet.js, automated multi-channel alerts (SMS via Twilio, Email via SendGrid, and in-app notifications) triggered by configurable thresholds, and an interactive dashboard with role-based access for Admins, Analysts, Emergency Teams, and Public users. Additional features include historical trend analysis with D3.js visualizations, PDF/CSV report generation, ML model performance monitoring, and 99.9% uptime with auto-scaling AWS cloud infrastructure.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv